Digital Forensics paper

Analysis of 3 PCAP files

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Note that Information contained in this document is for educational purposes.

Contents

1 INTRODUCTION

1.1 BACKGROUND

Within the modern age, technology can very often be critical to orchestrating crime and most notably organised crime (How criminals use new technology, 1998). As with crime scene investigations which aim to use forensics to identify evidence of a crime, digital forensics are used to gather evidence of activity which took place on a device and commonly interacting with the internet and identify if any crime has occurred relating to the suspect.

1.2 AIM

Within this investigation, a thorough forensics investigation will be conducted relating to three different packet capture (PCAP) files. The analyst had been tasked with unique objectives for each of the PCAP files while various digital forensics tools and practices are used to complete these objectives. These objectives all relate to obtaining evidence of potential drug trafficking the suspects have committed. The investigation conducted has been performed and documented in a method where it could be replicated by someone regardless of their technical ability.

1.3 METHODOLOGY

The methodology being employed for this investigation would be the one suggested by guru99 (Williams, 2023).

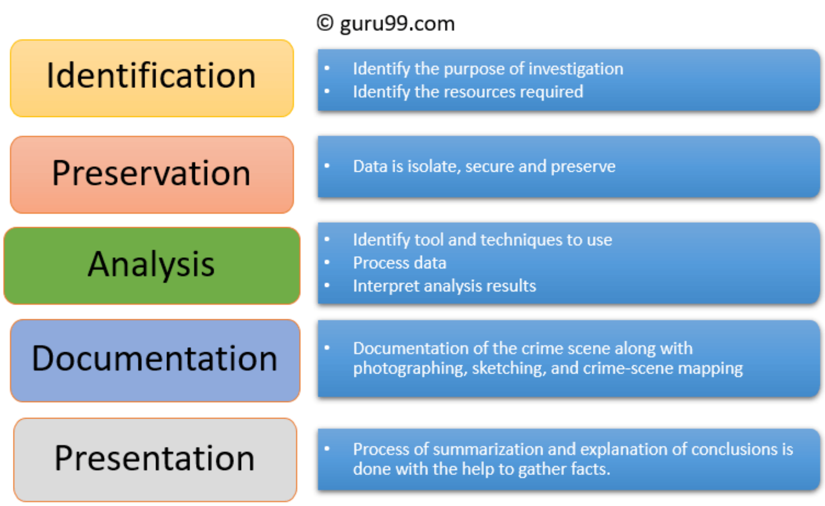


Figure 1-1 – Diagram showing the methodology employed by guru99.

This methodology employs a 5-step process to the digital forensics' procedure. Within this report, documentation will take place alongside the analysis to provide step by step instructions of how the data was acquired while the presentation is the report.

2 TOOLS USED WITHIN THIS INVESTIGATION

A list of tools which contributed to this investigation including a brief description can be found below.

Tool	Description	Used in PCAP
VMWare Workstation Pro (Kali VM)	Software used to launch and manage virtual machines. 'Kali' is a Linux distribution which comes with many forensics tool pre-installed.	1, 2, 3
Wireshark	A PCAP analysis tool used to display information about packets and export their contents.	1, 2, 3
Online URL decoder	An online tool used to decode URL encoded text.	1
Online Base64 decoder	An online tool used to decode Base64 encoded text.	1, 2
Microsoft Word	Software used for managing and viewing .docx files.	1
Engrampa Archive Manager	Software used for opening .zip files.	2
Google Chrome	Search engine used for viewing images.	2
HxD	A tool used for viewing the hexadecimal values of files.	2
Silent Eye	A steganography tool which can encode messages within images.	2
Google maps	An online web service which allows the user to identify	3

Table 2-1 – Table listing the tools used in the investigation.

3 PCAP 1 INVESTIGATION

3.1 IDENTIFICATION

To begin the investigation of PCAP 1, identification of the objectives took place. The investigator was tasked with locating files which relate to smuggling of illegal substances. This file should contain a record of the drugs and their quantities. Additionally, any methods of encoding the files should be identified alongside how to decode the files.

3.2 PRESERVATION

The necessary measures of verifying that the data (in this case, a single PCAP file) was performed with hashing, using MD5sum. Comparing the same hashes verifies the data integrity.

3.3 ANALYSIS

Due to the objective being to find files related to the drugs and quantities, identifying the network protocol used to share this file is critical. One of these protocols would be ‘SMB2’ which is a common network protocol used to share files between machines within a network (Sheldon, 2023). Wireshark, an extremely common tool used for network forensics and packet analysis has the capability of filtering out and displaying packets based upon a specified network protocol.

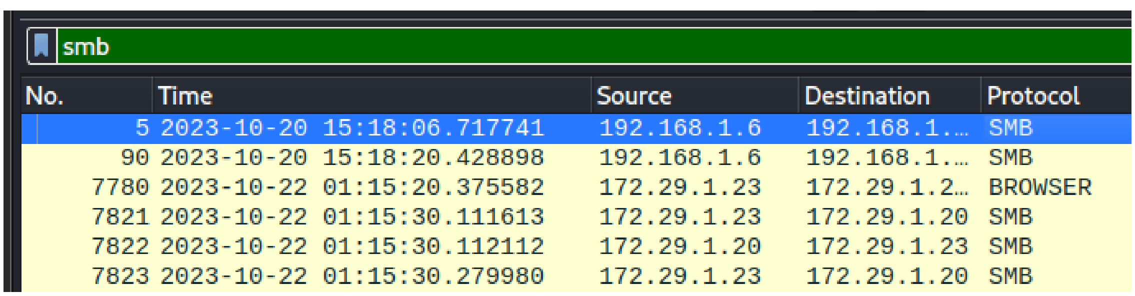


Figure 3-1 – Image displaying PCAP 1 being filtered in Wireshark for SMB packets.

From the filters applied, filesharing through the use of SMB has taken place. Wireshark also has the capability to export the contents of found within the PCAP files. This can be specified based upon the network protocol, such as HTTP and SMB. When exporting the contents received or sent via SMB, many files relating to the investigation are present. Within these files, numerous docx files can be found, which include “Substances” in the file name:

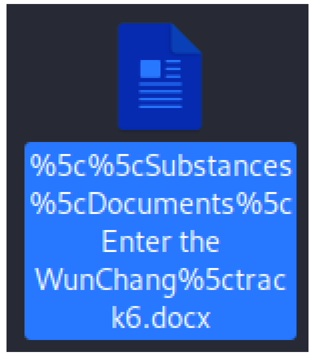


Figure 3-2 – Document found within PCAP 1.

Within the filename, URL encoding is used as %5c decodes to be a '/'. Using an online URL decoding tool, the filename translates to '\\Substances\Documents\Enter the WunChang\track6.docx', which gives hints to the suspects file directory. The contents of this file are shown below:

U2Vuc2l0aXZlIGluZm9ybWF0aW9uDQpEcnVncyBSZWNvcmRzCQkNCk51bWJlcgkJC
QlEcnVnIE5hbWUJCQkJQW1vdW50DQoxCQkJCQlBdG9ydmFzdGF0aW4JCQkxMTQ1
MDk4MTQNCjIJCQkJCUxldm90aHlyb3hpbmUJCQk5ODk3MDY0MA0KMwkJCQkJTW
V0Zm9ybWluCQkJCTkyNTkxNDg2DQo0CQkJCQlMaXNpbm9wcmlsCQkJCTg4NTk3M
DE3DQo1CQkJCQlBbWxvZGlwaW5lCQkJCTY5Nzg2Njg0DQo2CQkJCQlNZXRvcHJvbG
9sCQkJCTY2NDEzNjkyDQo3CQkJCQlBbGJ1dGVyb2wJCQkJNjE5NDgzNDcNCjgJCQkJC
U9tZXByYXpvbGUJCQkJNTYzMDAwNjQNCjkJCQkJCUxvc2FydGFuCQkJCTU0ODE1ND
ExDQoxMAkJCQkJR2FiYXBlbnRpbgkJCQk0OTk2MTA2Ng0KMTEJCQkJCUh5ZHJvY2h
sb3JvdGhpYXppZGUJCTQxNDc2MDk4

Figure 3-3 - Contents of track6.docx encoded in Base64.

The contents of track10.docx are believed to be encoded in base64. Using an online decoding tool, the true contents are revealed:

Sensitive information
Drugs Records

Number	Drug Name	Amount
1	Atorvastatin	114509814
2	Levothyroxine	98970640
3	Metformin	92591486
4	Lisinopril	88597017
5	Amlodipine	69786684
6	Metoprolol	66413692
7	Albuterol	61948347
8	Omeprazole	56300064
9	Losartan	54815411

Figure 3-4 – Decoded contents of track6.docx.

A “Drugs Records” can be identified including the record number, name of the drug and the amount.

With the name of the drug records document acquired, identifying the packets where the first instance it appeared can occur:

218	2023-10-20 15:18:20.636793	192.168.1.6	192.168.1.20	SMB2	354	Create Request File: %5cSubstances\Documents\Enter the WunChang
219	2023-10-20 15:18:20.643610	192.168.1.20	192.168.1.6	SMB2	211	Create Response File: %5cSubstances\Documents\Enter the WunChang

Figure 3-5 – Packets being used to create the directory where the drugs record was found.

Within figure 3-5, the directory where track6.docx, can be seen being created at 15:18:20 on 2023-10-23 as indicated by the Create Request File aspect of the packet. It can be assumed that 192.168.1.6 is the user’s computer and that 192.168.1.20 is some kind of data holder within their network.

1081	2023-10-20 15:18:22.116410	192.168.1.6	192.168.1.20	SMB2	378	Create Request File: %5cSubstances\Documents\Enter the WunChang\track6.docx
1082	2023-10-20 15:18:22.128519	192.168.1.20	192.168.1.6	SMB2	211	Create Response File: %5cSubstances\Documents\Enter the WunChang\track6.docx

Figure 3-6 - Packets being used to create the file where the drugs record was found.

Next, ‘track6.doc’ is created at 15:18:22 on 2023-10-20 from the same IP source and destination addresses.

4 PCAP 2 INVESTIGATION

4.1 IDENTIFICATION

Similar to PCAP 1, the objectives of PCAP 2 relate to suspected drug trafficking within FTP (File transfer protocol) traffic, the objective is to find evidence of an item which a foreign contact had received. Additionally, the analysts have been informed that anti-forensics practices have been used to hide the information sent and that an Obi -Wan Kenobi quote will help decipher the message.

4.2 PRESERVATION

The necessary measures of verifying that the data (in this case, a single PCAP file) was performed with hashing, using MD5sum. Comparing the same hashes verifies the data integrity. Additionally, when directly editing evidence (insert the figure name here later), backups of the images were saved.

4.3 ANALYSIS

As the identification implies, the wanted contents could be found within FTP traffic. As the name implies, FTP is a networking protocol used for the transfer of files. Using Wireshark, a list of all the FTP packets were displayed:

```
 78 Request: PORT 192,168,1,6,126,2
105 Response: 200 PORT command successful. Consider using PASV.
 71 Request: RETR d3arth.zip
124 Response: 150 Opening BINARY mode data connection for d3arth.zip (9160 bytes).
 78 Response: 226 Transfer complete.
 78 Request: PORT 192,168,1,6,126,4
105 Response: 200 PORT command successful. Consider using PASV.
 70 Request: RETR dr0id.zip
123 Response: 150 Opening BINARY mode data connection for dr0id.zip (8864 bytes).
 78 Response: 226 Transfer complete.
 78 Request: PORT 192,168,1,6,126,5
105 Response: 200 PORT command successful. Consider using PASV.
 60 Request: NLST
 93 Response: 150 Here comes the directory listing.
 78 Response: 226 Directory send OK.
 60 Request: QUIT
 68 Response: 221 Goodbye.
 74 Response: 220 (vsFTPd 3.0.3)
 68 Request: OPTS UTF8 ON
 80 Response: 200 Always in UTF8 mode.
 68 Request: USER ftpuser
 88 Response: 331 Please specify the password.
 69 Request: PASS starwars
 77 Response: 230 Login successful.
 65 Request: CWD files
 91 Response: 250 Directory successfully changed.
 79 Request: PORT 192,168,1,6,126,21
105 Response: 200 PORT command successful. Consider using PASV.
 60 Request: NLST
 93 Response: 150 Here comes the directory listing.
 78 Response: 226 Directory send OK.
 79 Request: PORT 192,168,1,6,126,23
105 Response: 200 PORT command successful. Consider using PASV.
 80 Request: RETR untitled folder.zip
133 Response: 150 Opening BINARY mode data connection for untitled folder.zip (1322 bytes).
 78 Response: 226 Transfer complete.
 60 Request: QUIT
 68 Response: 221 Goodbye.
```

Figure 4-1 – PCAP 2 being filtered for exclusively FTP packets.

When examining the FTP packets, numerous .zip files could be found which indicated that critical information could be found within them. These .zip files were all taken requested by 192.168.1.6 and sent by 192.168.1.20. These .zip files are named:

- 3v0ke.zip
- C0llec3t.zip
- D3arth.zip
- Dr0id.zip
- Untitled folder.zip

Additionally, the credentials for communicating with the FTP server can be identified, with the username being ftpuser and the password being starwars. Wireshark has the capabilities to save these .zip files using the following steps which in this example will be performed on coll3ct.zip.

- Identify the .zip file in the filtered capture packed traffic:

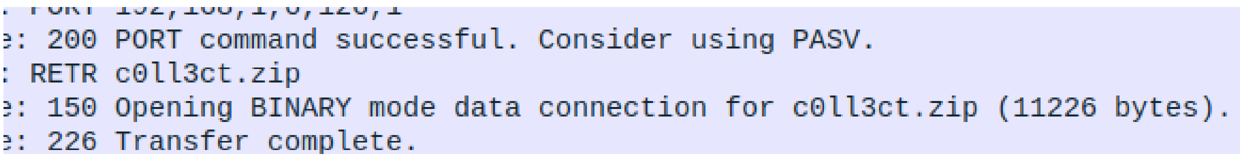


Figure 4-2 – Packet containing a zip file.

- Identify the RETR packet which is used for the file transfer of this .zip file.
    - In this case, it is frame 19287.
- Disable the FTP filter and identify the next FTP-data frame:

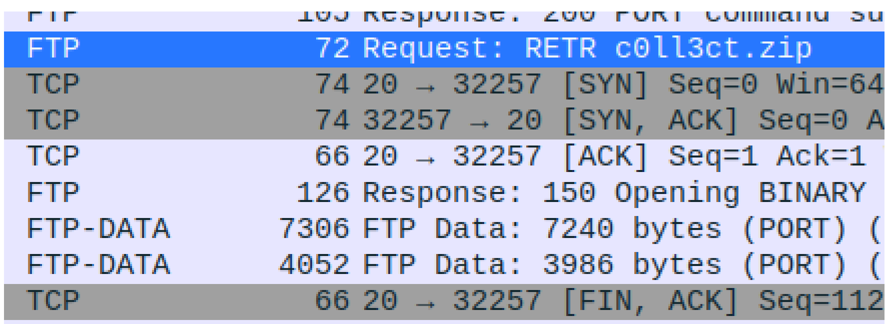


Figure 4-3 – Packets responsible for transferring the zip file, including the FTP-DATA.

- Right click on one of the FTP-data packets and view the TCP conversation.
- Save the TCP conversation as raw.

The steps were completed for each of the .zip files found through the FTP filter. Using Engrampa Archive Manager, the contents of each of the .zip files can be shown. The contents of the zip folders are shown below:

Name	Contents
C0llct3.zip	Battles.jpg, Can.jpg, I.jpg, Of.jpg
3v0ke.zip	How.jpg, Into.jpg, Never.jpg, Some.jpg
D3arth.zip	Game.jpg, These.jpg, Understand.jpg, You.jpg

Dr0id.zip	You.jpg, Kind.jpg, ll.jpg
Untitled folder.zip	Numerous empty folders until one is called 'Silent Eye'.

Table 4-1 – Contents of each of the zip files extracted from the PCAP.

From the zip folders, a total of 15 JPEG files can be found, all uniquely identified by a word. When attempting to view these JPEG images, all of them (with the exception of I.jpg) cause an error and are not viewable. When inspecting I.jpg, the image appears to look like a segment of an image:

Figure 4-4 – Image of I.jpg.

It can be assumed from this image that the reason the other JPEG images found within the other zip folders are appearing as an error is that they are all a part of one image and must be reconstructed into one. However, the order of which the images must be reconstructed is crucial as it will only correctly construct the image if done in the right order. Judging by how all the file names are common words, it can be assumed that this is where the hint for using an Obi Wan Kenobi quote comes into fruition. It can be assumed that by looking at the image, I.jpg is the first part of the image (and therefore 'I' is also the first word of the Obi Wan Kenobi quote) however to verify this, the image was viewed using the hex editor known as HxD:

```
Offset(h)  00 01 02 03 04
00000000   FF D8 FF E0 00
```

Figure 4-5 – First bytes of I.jpg.

When inspecting the hexadecimal values of I.jpg, the values known as the JPEG magic numbers are found. These hexadecimal values indicate that this is the beginning of completed image due to these values always appearing at the beginning of a JPEG image (Moore, 2015). Additionally, the file which should appear at the end of the image can be found using similar techniques:

```
00000A30   00 7F FF D9
```

Figure 4-6 – Last bytes of game.jpg.

The hexadecimal values ‘FF D9’ indicate the end of the JPEG image, ‘game.jpg’ (JPG Signature Format: Documentation & Recovery Example, 2023), suggesting that it is the last file to appear in the image but also that ‘game’ is the last word to appear the Obi Wan Kenobi quote.

Identifying the Obi Wan Kenobi quote was the next step in the investigation which was made easier due to knowing the beginning and end of the quote. Using an online source which contains 149 Obi Wan Kenobi quotes (Roychoudhury, 2021), a quote which contained all the image words was identified. This quote was: “I'll never understand how you can simplify these battles into some kind of game?”. With the order of the files identified, the files were combined using a command line tool known as cat (appendix B). The completed image resembled a ship.

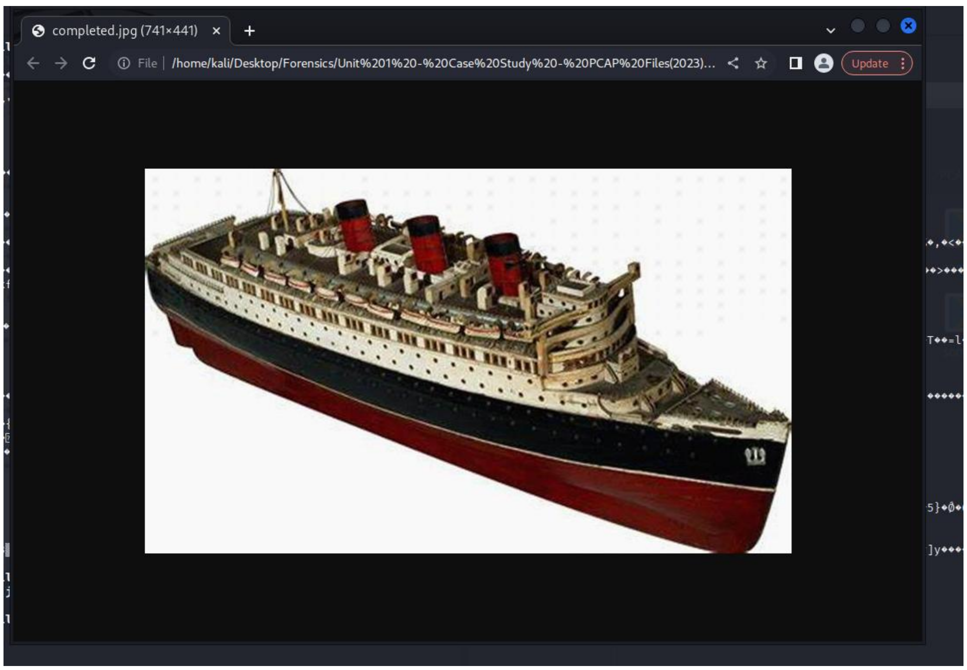


Figure 4-7 – Reconstructed image using the zip file contents.

The 5th zip folder contained various empty files until one of them was named “Silent Eye”. Upon research, Silent Eye is a steganography tool which can be used to hide data within images. With the current completed image from the zip folders, it can be heavily assumed that Silent Eye was used on this JPEG to encode a message inside of it. When using Silent Eye to decode the image, the following message is revealed:

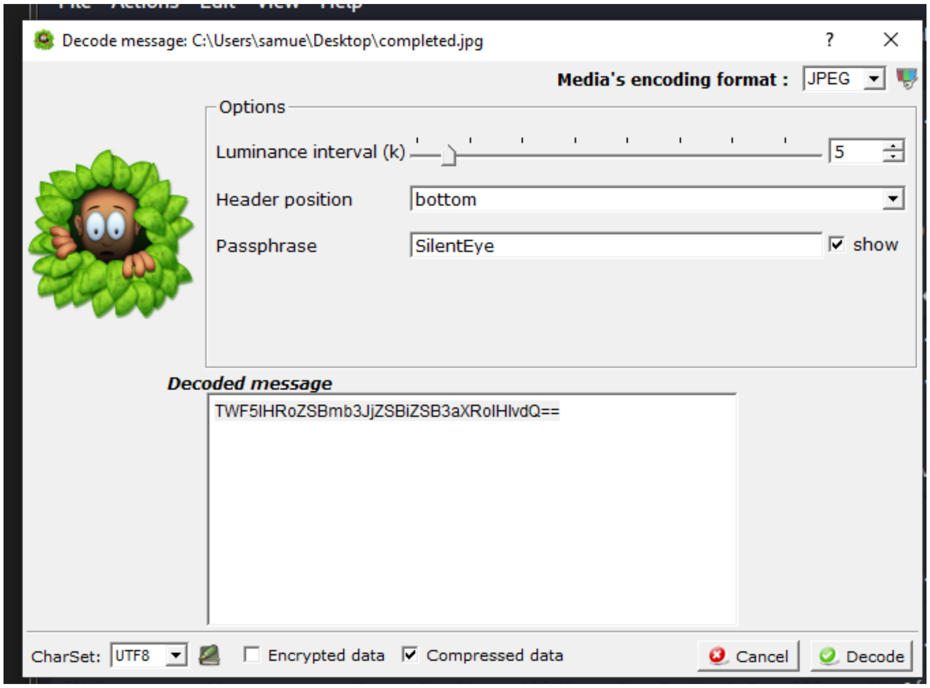


Figure 4- 8 – Decoded message found using Silent Eye.

The decoded message still appears to need to undergo further decoding as it appeared to be encoded using base64.

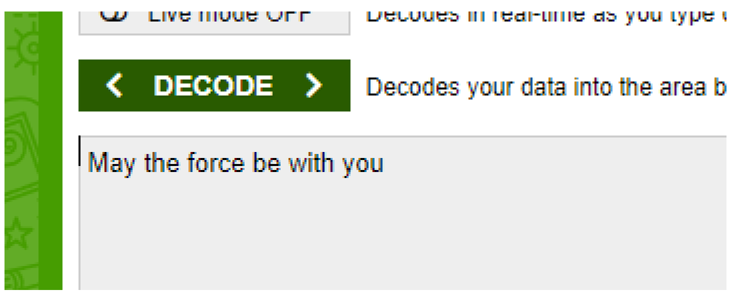


Figure 4-9 – Message decoded further from Base64.

The message is decoded again to reveal the famous Star Wars quote, ‘May the force be with you’.

5 PCAP 3 INVESTIGATION

5.1 IDENTIFICATION

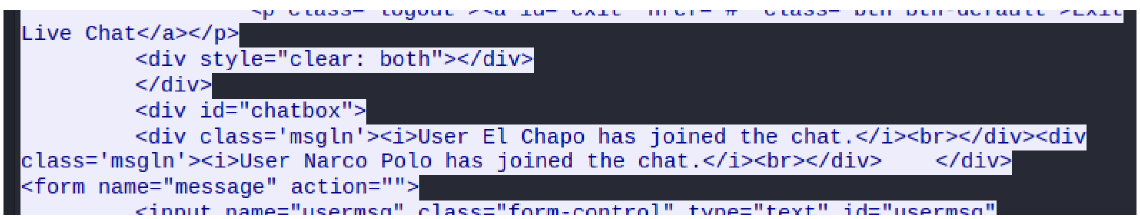
The objective for PCAP 3 involves identifying communication between El Chapo and a known drug trafficker known as Narco Polo. The analyst needs to find information relating to a secret meeting between them.

5.2 PRESERVATION

The necessary measures of verifying that the data (in this case, a single PCAP file) was performed with hashing, using MD5sum. Comparing the same hashes verifies the data integrity.

5.3 ANALYSIS

The analysis of PCAP 3 began in Wireshark. Filtering for HTTP packets, a conversation happening within TCP stream 234 can be seen. A TCP stream reliably sends segments of the data to the user to allow for effective and efficient routing (Transmission Control Protocol, n.d.). When viewing TCP stream 234, the HTTP packets used for an online messaging service are seen:



```
Live Chat</a></p>
        <div style="clear: both"></div>
        </div>
        <div id="chatbox">
        <div class='msgln'><i>User El Chapo has joined the chat.</i><br></div><div
class='msgln'><i>User Narco Polo has joined the chat.</i><br></div>    </div>
<form name="message" action="">
```

Figure 5-1 – HTTP packet contents showing the beginning of a conversation between El Chapo and Narco Polo.

Within this HTTP packet, it can be seen that a ‘chatbox’ between El Chapo and Narco Polo has been created. This conversation takes place with 192.168.1.6 between 192.168.1.20. Scrolling to the end of the TCP stream, the entire conversation can be observed:

```
<div class='msgln'><i>User El Chapo has joined the chat.</i><br></div><div
class='msgln'><i>User Narco Polo has joined the chat.</i><br></div><div
class='msgln'>(12:46 PM) <b>El Chapo</b>: Good evening, Narco Polo.<br></div><div
class='msgln'>(12:46 PM) <b>Narco Polo</b>: Who&#039;s on the line?<br></div><div
class='msgln'>(12:46 PM) <b>El Chapo</b>: Phoenix.<br></div><div class='msgln'>(12:46 PM)
<b>Narco Polo</b>: Where are you?<br></div><div class='msgln'>(12:46 PM) <b>El Chapo</b>:
I can&#039;t disclose that information, even to you.<br></div><div class='msgln'>(12:46
PM) <b>Narco Polo</b>: Are you aware of the current scrutiny on El Chapo?<br></div><div
class='msgln'>(12:47 PM) <b>El Chapo</b>: Yes, I&#039;m fully aware, However, they will
never know it is me behind the shipment.<br></div><div class='msgln'>(12:47 PM) <b>Narco
Polo</b>: Regardless, we must exercise the highest level of secrecy. Be vigilant.
I&#039;d like to meet in 2nd November at 10 PM to plan the secret delivery and avoid any
complications.<br></div><div class='msgln'>(12:47 PM) <b>El Chapo</b>: At our usual
rendezvous point?<br></div><div class='msgln'>(12:47 PM) <b>Narco Polo</b>: Yes<br></
div><div class='msgln'>(12:47 PM) <b>El Chapo</b>: What day?<br></div><div
class='msgln'>(12:47 PM) <b>Narco Polo</b>: I already mentioned, stay sharp.<br></div>
```

Figure 5-2 – Entire conversation recorded between El Chapo and Narco Polo.

Within this conversation numerous points of interest can be observed:

- Conversation took place between 12:46pm and 12:47pm.
- El Chapo appears to under the alias ‘Phoenix’.
    - From this it can be assumed that El Chapo is receiving this conversation.
    - Narco Polo may also not be aware that Phoenix is El Chapo.
        - This can be assumed in the way Narco Polo refers to ‘the current scrutiny on El Chapo’.
        - However, Narco Polo may be attempting to keep up with the Pheonix alias.
- Mentions discussion of the shipment.
- Narco Polo suggests meeting on the 2nd of November at 10pm at the usual rendezvous point.

Interestingly, in a later TCP conversation (TCP stream 247) Narco Polo can be seen revealing more information about the meeting:

```
div><div class='msgln'>(12:47 PM) <b>El Chapo</b>: What day?<br></div><div
class='msgln'>(12:47 PM) <b>Narco Polo</b>: I already mentioned, stay sharp.<br></
div><div class='msgln'>(12:53 PM) <b>Narco Polo</b>: 36.62575185817829
-117.08896804489794<br></div>
```

Figure 5-3 – Additional messages found in another HTTP stream including coordinates.

At 12:53pm, Narco Polo is seen revealing coordinates which can be assumed to be meeting point for the shipment. Using google maps, the coordinates lead to a place within Death Valley, California.

An obscurity within this PCAP would be the timestamps of the packets:

```
9918 2014-07-02 12:52:32.700124
9919 2014-07-02 12:52:32.749857
9920 2023-10-22 12:45:14.750286
```

Figure 5-4 – Timestamps for packets showing a large time gap.

As seen in figure 5-, between frame 9919 and 9920 there is an approximately 9-year gap. The packets found relating to the earlier conversation are found after the time gap. This time gap raises suspicions that this PCAP file may not be fully forensically sound.

6 Critical Evaluation

6.1 PCAP 1

Wireshark played a critical role in the file acquisition aspect of the investigation into PCAP 1. Additionally, being able to filter out specific packets based upon their network protocol, such as SMB or TCP, or identifying specific TCP streams greatly contributed to gaining context to what was happening in the packet capture.

When the SMB files had been extracted, very minor data obfuscation and encoding had taken place. Technically, the file naming of the drug record document not relating to anything to do with would count as an anti-forensics practice. While there is URL encoding within the name of the document, it is suspected that this is not intentionally done for data obfuscation purposes but is just automatically done during packet transfer and is only noticeable due to the files coming from a directory.

Within the file itself, base64 is used to completely hide the table of drug records. Base64 encoding is very recognisable and most experienced network forensics should be able to identify it either from trial and error or just looking at it, making it a weak anti-forensics practice if the file is directly found. However, the base64 encoding is effective in preventing some network forensics tools. For example, using the tool known as grep to search in the SMB file shares for keywords such as drugs would not work due to the base64 encoding.

6.2 PCAP 2

To begin the investigation of the second PCAP file, the filtering tool on Wireshark was used again, this time for packets sent using the FTP network protocol. Wireshark proved to be very effective in identifying the exact packets where each of the zip files containing sensitive information were being held and subsequently extracting them via the filtered TCP stream.

Within the zip files, the deconstructed image of a boat was found. This image was split across 4 zip files and 15 images resulting in the analyst being required to attempt to reconstruct the image. This was an effective anti-forensics method and would have taken much more time to reconstruct and solve if the hint relating to the Obi Wan Kenobi quote had not been given. Conversely, it was very easy to tell that something was suspicious about the contents of the zip folders from their random yet purposeful names to that the beginning and end of the completed JPEG image could be identified by observing the hexadecimal values of the images.

Even when the image had been reconstructed, further anti-forensics practices had been used. Within the only zip folder which did not have a Star Wars or ‘L33T’ name was an empty file known as Silent Eye. This was quickly identified to be a steganography tool suggesting that it had been used on this already suspicious image. Upon using the decode feature of Silent Eye, a string encoded in Base64 was revealed. This Base64 encoding served as an incredibly weak anti-forensics tool, especially in this context. When during the PCAP 1 investigation, the Base64 encoding prevented keywords from being searched, this would not occur in the context of a message being hidden inside an image due to steganography. While the message found did not relate to an item a foreign contact had received, no further information could be found from the PCAP file.

6.3 PCAP 3

The investigation of PCAP 3 took place entirely using Wireshark and viewing the contents of the TCP stream where the conversation took place. This allowed for the conversion between El Chapo and Narco Polo to be observed. Within this conversation, there were 0 attempts at data obfuscation, anti-forensics practices or encoding. Due to this, the information required for the investigation is extremely visible and suspiciously easy to find.

An issue within PCAP 3 is that there is a large time skip of roughly 9 years which could suggest that the evidence was tampered with, thus the PCAP cannot confidently be recognised as forensically sound.

7 Reflective component

The methods a malicious hacker could use digital forensics to impede, obscure or even stop an investigation are numerous. These methods mainly revolve around the use of using tools to perform data obfuscation which aims to make identification of critical and sensitive information much harder to find within an investigation. Commonly, this is will greatly slow down the investigation as the most effective way of countering data obfuscation and anti-forensic practices is to identify what method they used and utilise a tool to counteract this obfuscation.

Within PCAP 1 and 2, numerous data obfuscation techniques such as Base64 encoding and file naming were used which did succeed in slowing down the investigation, however Base64 encoding was considerably more effectively used within PCAP 1. PCAP 2 by far utilised the greatest anti-forensics techniques, not only relying on an image to be reconstructed based around a (relatively obscure) Obi Wan Kenobi quote, but also said image undergoing steganography. These techniques combined are incredibly effective at slowing down an investigation due to obscuring the information and the image reconstruction not being solved with just a simple tool. PCAP 3 conversely utilised 0 data obfuscation techniques resulting in the information being extremely easy to acquire. This was almost suspiciously too easy to obtain. That combined with the odd timestamps raises the idea that the PCAP file may not be forensically sound.

If a malicious actor wanted to use more anti-forensics techniques to further complicate an investigation, they should consider using other network protocols for communication, such as Transport Layer Security (TLS). TLS provides end-to-end encryption for other network protocols such as HTTPS which aims to protect the data sent and received by a user (Cloudflare, n.d.).

8 References

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APPENDICES

APPENDIX A – RECOVERED DOCUMENTS RELEVANT TO PCAP 1 INVESTIGATION

7.1.1 Track6.docx

Sensitive information

Drugs Records

Number	Drug Name	Amount
1	Atorvastatin	114509814
2	Levothyroxine	98970640
3	Metformin	92591486
4	Lisinopril	88597017
5	Amlodipine	69786684
6	Metoprolol	66413692
7	Albuterol	61948347
8	Omeprazole	56300064
9	Losartan	54815411
10	Gabapentin	49961066
11	Hydrochlorothiazide	41476098

7.2 APPENDIX B – RECOVERED DOCUMENTS RELEVANT TO PCAP 2 INVESTIGATION

7.2.1 Hex dump for I.jpg

FF D8 FF E0 00 10 4A 46 49 46 00 01 01 01 00 78 00 78 00 00 FF DB 00 43 00 08 06 06 07 06 05 08 07 07
07 09 09 08 0A 0C 14 0D 0C 0B 0B 0C 19 12 13 0F 14 1D 1A 1F 1E 1D 1A 1C 1C 20 24 2E 27 20 22 2C 23
1C 1C 28 37 29 2C 30 31 34 34 34 1F 27 39 3D 38 32 3C 2E 33 34 32 FF DB 00 43 01 09 09 09 0C 0B 0C 18
0D 0D 18 32 21 1C 21 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32
32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 FF C0 00 11 08 01 B9 02 E5 03 01
22 00 02 11 01 03 11 01 FF C4 00 1F 00 00 01 05 01 01 01 01 01 01 00 00 00 00 00 00 00 00 01 02 03 04
05 06 07 08 09 0A 0B FF C4 00 B5 10 00 02 01 03 03 02 04 03 05 05 04 04 00 00 01 7D 01 02 03 00 04 11
05 12 21 31 41 06 13 51 61 07 22 71 14 32 81 91 A1 08 23 42 B1 C1 15 52 D1 F0 24 33 62 72 82 09 0A 16
17 18 19 1A 25 26 27 28 29 2A 34 35 36 37 38 39 3A 43 44 45 46 47 48 49 4A 53 54 55 56 57 58 59 5A 63
64 65 66 67 68 69 6A 73 74 75 76 77 78 79 7A 83 84 85 86 87 88 89 8A 92 93 94 95 96 97 98 99 9A A2 A3
A4 A5 A6 A7 A8 A9 AA B2 B3 B4 B5 B6 B7 B8 B9 BA C2 C3 C4 C5 C6 C7 C8 C9 CA D2 D3 D4 D5 D6 D7 D8
D9 DA E1 E2 E3 E4 E5 E6 E7 E8 E9 EA F1 F2 F3 F4 F5 F6 F7 F8 F9 FA FF C4 00 1F 01 00 03 01 01 01 01 01
01 01 01 01 00 00 00 00 00 00 01 02 03 04 05 06 07 08 09 0A 0B FF C4 00 B5 11 00 02 01 02 04 04 03 04
07 05 04 04 00 01 02 77 00 01 02 03 11 04 05 21 31 06 12 41 51 07 61 71 13 22 32 81 08 14 42 91 A1 B1
C1 09 23 33 52 F0 15 62 72 D1 0A 16 24 34 E1 25 F1 17 18 19 1A 26 27 28 29 2A 35 36 37 38 39 3A 43 44
45 46 47 48 49 4A 53 54 55 56 57 58 59 5A 63 64 65 66 67 68 69 6A 73 74 75 76 77 78 79 7A 82 83 84 85
86 87 88 89 8A 92 93 94 95 96 97 98 99 9A A2 A3 A4 A5 A6 A7 A8 A9 AA B2 B3 B4 B5 B6 B7 B8 B9 BA C2
C3 C4 C5 C6 C7 C8 C9 CA D2 D3 D4 D5 D6 D7 D8 D9 DA E2 E3 E4 E5 E6 E7 E8 E9 EA F2 F3 F4 F5 F6 F7 F8
F9 FA FF DA 00 0C 03 01 00 02 11 03 11 00 3F 00 F7 9A 28 A8 7E 6D FF 00 EC D0 04 D4 51 45 00 14 52 77
A5 A0 02 8A 2A 1F 9B 7F FB 34 01 35 14 51 40 05 15 0F CD BF FD 9A 9A 80 0A 28 A4 EF 40 0B 41 A2 83

40 05 31 80 3C 13 D7 A6 7D 69 D4 87 19 1C 67 9F CA 8E 81 D4 F2 1F 88 9E 20 95 FC 67 61 A0 BD B2 B4
42 23 27 9A BF C2 C4 62 AE F8 22 F2 7D 03 58 4D 1E F9 23 29 78 BF B9 9E 0F B8 18 72 54 E7 BD 68 78 97
4A B5 D4 2E F5 77 68 14 CC B0 21 57 EE 08 3D AA 2D 0A DB FB 6B C1 B7 11 46 15 6F 6D 27 32 46 7F 88
11 CE 3F 1A C5 7C 66 BD 0F 47 5E 94 B5 97 A1 EA 83 55 D2 62 B9 C7 EF 71 B6 45 1D 9C 75 15 A4 18 11
91 5A AD CC 87 51 49 DE 96 98 05 14 52 77 A0 05 A2 8A 28 00 A2 93 BD 2D 00 14 51 50 FC DB FF 00 D9
A0 09 A8 A2 8A 00 28 A4 EF 4B 40 05 14 52 77 A0 05 A2 8A 28 00 A2 93 BD 2D 00 14 51 49 DE 80 16 8A
28 A0 02 8A 4E F4 B4 00 51 45 43 F3 6F FF 00 66 80 26 A2 8A 28 00 A2 A1 F9 B7 FF 00 B3 53 50 01 45 15
0F CD BF FD 9A 00 9A 8A 28 A0 02 8A 87 E6 DF FE CD 4D 40 05 14 52 77 A0 05 A2 8A 28 00 A2 A1 F9 B7
FF 00 B3 53 50 01 45 15 0F CD BF FD 9A 00 9A 8A 28 A0 02 8A 4E F4 B4 00 51 45 27 7A 00 5A 28 A2 80
0A 2A 1F 9B 7F FB 35 35 00 14 51 50 FC DB FF 00 D9 A0 09 A8 A2 8A 00 28 A8 7E 6D FF 00 EC D4 D4 00
51 45 43 F3 6F FF 00 66 80 26 A2 8A 28 00 A2 93 BD 2D 00 14 51 50 FC DB FF 00 D9 A0 09 A8 A2 8A 00
28 A2 A1 F9 B7 FF 00 B3 40 13 51 45 14 00 51 50 FC DB FF 00 D9 A9 A8 00 A2 8A 4E F4 00 B4 51 45 00 14
54 3F 36 FF 00 F6 6A 6A 00 28 A2 93 BD 00 2D 06 8A 28 00 A6 9C 8D C7 DA A3 F9 B7 FF 00 B3 4F 73 FB
B7 CF A1 A0 0C 0B 78 16 6D 52 F4 B8 C8 96 2E 95 CA 7C 3B B9 30 F8 AF 5F D3 58 E3 61 12 2E 7D CD 76
3A 79 FF 00 4B 70 46 4F 93 FD 6B C9 FC 2F A8 B4 3F 1A 6F 6D C9 3B 25 CA 91 DB 8E 95 8E D2 34 5A A3
D0 A4 93 FE 11 8F 16 2E EF 93 4C D4 49 E4 F4 8E 5F FE BD 75 EA 06 07 B5 65 F8 87 46 8F 5C D1 A6 B2 93
86 71 BA 36 3F C0 E3 A1 15 9F E1 1D 56 7B 8B 37 D3 F5 03 8B FB 33 E5 3E 7A B8 1D 0D 68 9E A6 67 4F
45 26 47 E2 69 6A 80 28 A2 93 BD 00 2D 14 51 48 02 8A 6E 4F 63 46 79 F7 FA D3 01 D4 53 72 45 47 96 CF
27 83 40 13 51 48 3D 0F 6E F4 B4 00 51 50 FC DB FF 00 D9 A9 A8 00 A2 8A 87 E6 DF FE CD 00 4D 45 14
50 01 45 43 F3 6F FF 00 66 A6 A0 02 8A 29 3B D0 02 D1 45 14 00 51 50 FC DB FF 00 D9 A9 A8 00 A2 8A
87 E6 DF FE CD 00 4D 45 14 50 01 45 27 7A 5A 00 28 A2 93 BD 00 2D 14 51 40 05 15 0F CD BF FD 9A 9A
80 0A 28 A8 7E 6D FF 00 EC D0 04 D4 51 45 00 14 54 3F 36 FF 00 F6 6A 6A 00 28 A2 93 BD 00 2D 14 51
40 05 15 0F CD BF FD 9A 9A 80 0A 28 A4 EF 40 0B 45 14 50 01 45 43 F3 6F FF 00 66 A6 A0 02 8A 29 3B D0
02 D1 45 14 00 51 49 DE 96 80 0A 28 A4 EF 40 0B 45 14 50 01 45 43 F3 6F FF 00 66 A6 A0 02 8A 2A 1F 9B
7F FB 34 01 35 14 51 40 05 14 54 3F 36 FF 00 F6 68 02 6A 28 A2 80 0A 2A 1F 9B 7F FB 35 35 00 14 51 49
DE 80 16 8A 28 A0 02 8A 4E F4 B4 00 51 45 43 F3 6F FF 00 66 80 26 A2 8A 28 00 AA B7 F2 18 ED 24 61 C7
41 FA D5 9E F5 43 53 21 9A DE 12 78 77 E4 7B 0E 69 30 2A D9 2E 35 0B CF 55 8F 03 15 E2 52 41 3E 99 F1
4A 3B F6 0C 8B 3C DB 54 91 CF 5E 6B DC 74 96 05 27 BA 7C 27 9B 21 C6 7D 2B 97 F1 FC 16 97 30 5B 4F
6A 23 F3 E3 90 EF 61 F7 97 02 B0 AB 74 B9 8D 29 EF 63 B8 53 BD 43 83 B8 1E 40 AE 7B 5C D2 E4 B7 BC
4D 6B 4F 52 B7 11 FF 00 AD 51 FC 6B DF F1 A9 F4 3D 6E C2 7D 3A DA 23 76 86 64 88 6E C9 E7 38 AD 68
EF AD 26 18 59 E3 60 4E 31 9A D6 0F DD B9 0F 46 45 A6 EA 71 6A 56 2B 73 17 3D 99 7B A9 F7 AB C0 83
C8 AE 4A FA CA E7 44 BF 7D 57 4E 5F 32 16 FF 00 5F 6E 0F 51 EA 2B A4 B1 BB 82 F6 D5 26 81 C3 07 19 C7
71 ED 54 22 D5 37 70 FE 94 A3 38 24 8A C5 D5 75 EB 3D 25 BC B9 98 B4 AC BB 96 31 D4 D1 A7 51 A3 6B
70 CE 2A 39 AE 61 B7 81 A6 9A 54 8E 25 19 67 63 80 3F 1A CF B1 D5 E0 D4 6D 64 95 3E 52 83 2F 1E 79 02
BE 70 F8 AD E2 BD 5F 53 D6 DA 11 7A EB A7 29 2B 1C 11 9D A3 F1 C7 5A 98 B8 B6 16 3D A3 57 F8 B9 E1
2D 25 99 1A F4 DC 38 38 C4 0B B8 1F C6 B9 5B BF 8F 16 FF 00 37 D8 34 87 90 76 69 0E DA F9 FA 2B 85 89
F2 63 00 FA D5 FF 00 3D FC AD CB 93 9E FD A9 FB C5 AE 43 D4 EE 3E 3C EB 7B CF 97 A6 C3 18 ED 96 CD
75 3F 0E 7E 2C 5D 78 AB 5D 3A 46 A5 6F 1C 72 BA 17 8D 93 A1 02 BE 78 77 79 5B 0C 71 5B 3E 0C D4 1B
45 F1 A6 9B 78 A4 ED 59 94 39 FF 00 64 F5 A5 76 B7 14 B9 6D A1 F6 40 23 A0 A7 53 23 90 4B 12 48 BF 75
C0 61 F4 34 FA A4 C8 0A 2A 1F 9B 7F FB 35 35 30 0A 28 A4 EF 40 0B 45 14 50 01 45 27 7A 5A 00 28 A2 A1
F9 B7 FF 00 B3 40 13 51 45 14 00 51 50 FC DB FF 00 D9 A9 A8 00 A2 8A 87 E6 DF FE CD 00 4D 45 14 50 01
45 27 7A 5A 00 28 A2 93 BD 00 2D 14 51 40 05 14 9D E9 68 00 A2 8A 87 E6 DF FE CD 00 4D 45 14 50 01
45 43 F3 6F FF 00 66 A6 A0 02 8A 2A 1F 9B 7F FB 34 01 35 14 51 40 05 14 9D E9 68 00 A2 8A 87 E6 DF FE
CD 00 4D 45 14 50 01 45 27 7A 5A 00 28 A2 93 BD 00 2D 14 51 40 05 15 0F CD BF FD 9A 9A 80 0A 28 A8

7E 6D FF 00 EC D0 04 D4 51 45 00 14 54 3F 36 FF 00 F6 6A 6A 00 28 A2 93 BD 00 2D 14 51 40 05 14 52 77
A0 05 A2 8A 28 00 A2 9A 4E 3A D3 B2 3D 68 00 A2 93 23 A6 68 C8 27 AD 2B 00 B4 C7 7D B9 27 A0 F5 E9
4E 66 55 19 62 00 F7 AC 5D 43 C4 9A 75 A2 AF EF 44 CC E0 ED 08 7A E2 86 EC 16 36 37 37 39 18 C7 A5 43
73 7B 6F 68 8C F3 CC 88 14 65 B2 7A 0A F3 6D 63 E2 5B B0 48 EC 00 0D 22 1C F9 5C BA 37 6C F6 AE 22 FB
55 D6 B5 09 E0 7B DB AF 2E 5F 2C AB B2 9C 99 54 F6 61 D0 1A 89 54 49 0D 46 EC F6 79 FC 65 A4 5B EA
31 59 CB 76 8A 65 50 51 81 E0 E7 A5 6F 2B EE 00 A9 DC 31 9C 8E F5 F3 D6 85 A2 DB EA 9A F5 95 93 3E E2
18 38 66 6E 54 0E D5 EC BA 87 89 F4 BD 16 DE 54 92 60 5A D9 06 54 1E 49 C7 4A 51 A9 71 B8 D9 9D 0E
58 F2 78 E6 AA DC EA 76 D6 99 13 4C 88 E0 6E 08 4F 24 7A D7 95 EB 5F 16 62 96 46 87 4E 47 91 26 88 8D
A4 63 63 7D 6B CF B5 6F 14 6A 57 5E 5F DA EF 16 24 08 54 A0 6C B7 E7 54 E7 D8 56 3E 8B 9B C4 7A 65
BC 51 BB DD 47 97 FB AA 0D 71 DA F7 C4 1B 2B 6B 99 25 8E 48 D7 EC F9 5C 39 FB C4 F7 15 E0 72 6B 97
1E 6F 97 04 92 BE 07 73 FA D5 69 5A 69 59 25 B8 62 C8 4E 01 63 D4 D4 3A 8D 21 A8 DD 9E AD 75 E3 4D
59 B4 66 BB 68 BC BB 75 93 74 4D 21 DB 9F 4C 7A D7 37 A1 78 86 FB 59 F1 6C B2 DF CB B9 E6 5E C7 E5
C7 D2 B4 3C 63 A7 4B 7D E0 BD 11 95 B6 A4 0A 19 80 E8 DF 5A E2 ED 75 08 74 EB E8 65 87 69 63 28 07
1D 87 A5 63 29 39 C1 A3 64 94 59 AB 79 A9 5C E8 BA F5 C9 B6 32 3F 9A DD 33 F7 6A 45 F1 7E A5 6F 97
D9 22 46 AD 9E 09 C9 35 1F 8D 50 C3 AC C3 3D BB 00 2E 23 0C 7D AA 7F 04 47 0E B1 E2 05 8A F9 33 12
29 2A A7 F8 88 EF 44 1B 54 75 14 D2 65 EB 6F 1F EB B6 4A B2 C5 2D C0 45 3B B7 48 B9 1B 4F 51 CD 6F 78
5F C5 7A 84 9A AF DA B4 99 73 64 01 7B A8 64 E9 9E A4 83 5C AF 8A 4E A1 2E BD 73 A4 B9 48 74 F5 4C
A3 28 C6 47 A5 3B C1 17 11 DB DF 9D 3D 1B 2B 3C 64 6D EF 91 D0 D3 94 DD 87 1A 49 6E 76 A7 E3 3D DC
D3 CD 15 AD 9A B2 DB 31 DE CF C1 7E 7A 01 52 5B F8 E6 1F 14 EA 0A B3 5A A5 BD C2 47 85 71 C9 AF 34
F1 36 9F F6 2B 59 A6 C1 49 FC EF DE 6D E3 8F 6A A1 A0 5C DC 59 5E 2D EB 36 D4 61 95 07 AE 3D 69 F3
73 44 4E 29 35 63 D6 34 8F 10 A6 85 A8 5E B5 E4 CD B6 E2 36 4E 7D 71 C1 AF 25 F1 2E C9 AC 96 42 FB
DD 1C 96 3D F2 4D 5B D5 AF A7 BA 96 69 4C 85 B2 72 A7 B6 2B 1D 27 F3 A2 9D 67 52 51 86 49 F4 35 14
DD 98 4A 37 7A 1C F5 4E 93 EC 50 37 13 ED 57 05 82 48 E5 80 20 1E 70 7B 54 D6 F6 68 88 77 A8 23 3C 57
4F B7 48 DA 38 0A

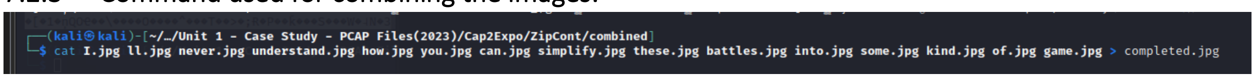
7.2.2 Hex dump for battles.jpg

71 E0 E2 BA 9D 77 E1 FF 00 88 FC 3E 4F 9F 60 F2 C2 3F E5 AC 2B BE B9 76 0E 87 6B 2E D7 F4 3F 7B F2 AC
9C 5A 3B E1 5A 0D 68 C6 1C 8E A2 93 3E A7 35 29 50 58 03 9C FD 72 69 36 E1 B9 C7 E1 48 AB 37 AA 19
82 DD 45 2A 82 AD F7 88 FA 52 90 09 E0 91 46 0F A6 68 B8 7B BB 3D C9 92 EA 78 CF CA FB 87 BD 58 8E
FB 2D FB D4 C7 B8 AA 43 1E 94 E0 0F A6 69 5C 89 53 4F 63 49 66 8A 4F E2 C5 2E DC F4 20 AD 66 F3 F4
A9 23 76 1D 1B 9A 69 DC CE 54 AD B1 71 A3 43 D4 53 7C 98 EA 25 9C F7 E6 A5 F3 94 80 08 C5 56 86 2E
E8 4F 2A 3F 53 4D 68 97 B3 54 EA AA CB 95 22 A3 68 46 7E 6C 93 48 B8 CC 88 42 70 72 7A 54 4E 30 38 E4
D4 ED 13 75 DF 8F 6A 89 86 7A 83 F5 A9 6A C6 AA A4 BB 91 F3 B7 A5 26 54 D3 8A 03 DC 8A 69 41 49 5C
B8 C9 B1 08 07 A7 34 6C 6F EE 9A 02 81 DC 8F A5 48 B2 48 BF 74 FE 74 6A 57 34 BB 11 10 47 04 60 D3 08
CF 18 A9 59 8B BE E7 1C 8E B4 8C 57 77 03 93 41 5C CC 88 E0 F6 E9 49 9C 8E 0E 05 4C 8E 10 10 50 1A 61
68 CF 24 6D 3E 95 48 CD B2 3E 3B 9A 42 A0 F4 A7 1D BD B3 49 8F 7A 62 B5 C6 80 54 1D BD 69 36 8C 60
74 3D 69 E1 37 77 A5 31 E3 A3 53 B9 3C 89 B1 A8 F2 46 7F 77 2B 27 FB A7 15 69 35 8D 4A 20 15 2F AE 17
1D C3 D5 32 29 BD 3B 50 44 A0 91 B6 9E 30 D7 E0 4F 2D 75 69 8A FA 13 9A 9A 2F 1D F8 86 00 A5 AF 5A
43 9F 94 1E 9F 8D 73 44 92 DC 8A D6 F0 FE 98 FA B6 BD 65 64 AB BB CD 99 43 0F 6C D6 91 48 E3 AD 64
8F A7 3E 19 69 F7 16 9E 13 8A E6 F0 93 75 76 4C B2 7A 73 D3 15 D9 D4 16 96 EB 69 69 05 B2 0F 96 28 C2
0F C0 62 A7 AD 92 B1 C0 DD C2 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0
02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02
8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A
28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 8A 28 A0 02 93 18 E9 4B 45 00 34 80 4F 7A
36 82 31 FC A9 D4 50 03 4C 6A 54 A9 19 53 D4 1E 6B 98 D7 7C 03 E1 CD 7A 32 2E F4 E8 D5 8F F1 C4 36

37 E6 2B A9 A4 34 9A 1A 93 5A A3 C2 F5 CF 81 33 C4 5A 4D 0E FC 32 1E 44 52 F0 47 FC 0A BC DF 58 F0
76 BF A0 C8 45 EE 9D 30 51 FF 00 2D 23 1B 97 F3 AF AE B9 E8 30 4D 32 58 62 99 3C B9 63 59 14 F5 0C B9
15 9B A7 73 A6 9E 2E 71 DC F8 B3 27 71 0D DB DA 9F 83 8C A1 C8 AF A8 B5 AF 86 7E 1A D6 83 3C B6 2B
14 DC E2 48 B8 C1 FA 57 9D EB 7F 04 2F 60 0C FA 4D F2 CE 3A 85 90 6D 22 A1 D2 67 64 31 94 DF C5 B9
E4 41 58 8F BB F9 D3 C0 20 F2 45 6A 6A DE 17 D6 B4 57 22 F7 4E B8 40 0E 37 60 E0 FB D6 51 2D 8E 57 68
F7 EB 59 38 F4 3A 7D AD F5 42 E3 3D 45 26 C5 EA 38 34 E0 E0 8C 52 52 E5 B1 A2 69 A1 17 2B DF 9F 5A
52 C7 3C D2 81 E9 49 DF 91 41 0D 46 F6 63 83 F3 95 38 A9 92 EA 45 C0 24 11 F4 A8 08 F4 14 98 3D A9 F3
58 52 A7 1E 85 D3 24 4F D7 83 48 54 9E 8C 31 55 73 FD E1 83 4E 12 15 EF C5 35 23 1F 67 34 48 E8 47 BD
46 47 B5 4A 92 29 FB C6 9E 42 1A AD C5 79 A2 AB 60 73 8A 8C AE EF 6A B4 D1 E4 70 71 51 B4 7E A3 3F
4A 96 8B 8D 49 10 9C 85 DB 9A 6F 60 31 D2 A4 23 1C 53 3A 52 B1 7E D2 E3 0F D2 93 20 9E 45 3C E2 9A
D8 3C 7A 53 48 AE 6B 8C E3 B9 A4 E2 95 80 23 8E D5 1E 0D 31 39 58 71 38 E9 40 3E A6 9B 49 9A 76 12
96 A3 CE 29 BD 39 CD 34 B5 20 39 ED 9A 12 B0 A5 2B 88 4E 4E 45 7A 9F C0 FD 14 5E F8 AE 4B D7 4C A5
AA 12 73 D3 27 A5 79 62 0C C9 8C ED AF A6 BE 0D 68 43 4B F0 80 BB 61 FB EB B6 DC C7 D8 74 AD 23 B9
E7 62 25 D0 F4 75 24 F5 E2 9D 4D 19 DC 46 38 1D 29 D5 B1 C6 14 51 45 00 14 51 45 00 14 51 45 00 14 51
45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00
14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51
45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00 14 51 45 00
14 51 45 00 14 51 45 00 18 1E 94 51 45 00 14 7E 14 51 40 11 4B 6D 0D C2 15 9E 24 95 7D 1D 41 AE 3F 5D
F8 61 E1 AD 6C 33 3D A0 82 52 7E FC 7C 7E 95 DA 11 9A 30 3D 29 35 71 A9 35 B1 E0 9A DF C1 1D 4E D0
3C 9A 4D D2 5C C7 D4 47 2F 0D F8 57 9E 6A 9E 1A D6 F4 56 2B A8 58 4F 06 3A 3B 2F 07 F2 AF AF B0 7D
73 51 4D 6B 0D C2 14 9A 14 91 4F 50 EA 0F F3 AC E5 4E E7 44 31 33 89 F1 90 71 9D BC 86 EF EF 4F 24 1E
02 9C FB D7 D3 7A DF C2 EF 0D 6B 20 9F B2 0B 59 0F F1 C3 C1 AF 3A D6 7E 06 EA 10 B3 C9 A4 DF 24 A8
3A 24 9D 4D 47 B2 3A 61 8C 4F E2 3C A0 6E 1E 94 64 77 AD 9D 57 C1 FA F6 88 EC 2F 74 D9 E3 40 33 E6
05 C8 3F 95 62 64 AF 6F 6E 78 AC A5 1B 1D 50 AB 09 0F A4 24 0E A2 97 B7 BF A5 00 8E 9F C4 7B 52 D8
DD 4A E3 39 ED 42 97 8D 32 0E 69 C5 41 38 3C 37 A5 07 8E 00 24 50 A5 62 25 01 E9 38 64 CB D3 B2 A5
72 87 93 55 D9 09 07 6F 7F 5A 00 21 00 19 C8 AB 52 31 E5 D4 97 07 BD 34 81 E9 9A 43 21 CE 0D 3C 15
E9 C8 3E B5 24 B5 62 02 BD 7B 54 27 83 D7 35 70 AF CB BB EF 54 0D 18 27 9C 03 E9 54 17 B1 0E EF 4A
8D 89 EF 52 98 8E 7A 53 4C 67 B5 31 36 D9 1E E1 DA 97 19 A5 23 D4 62 90 E4 50 52 D8 4A 51 C6 69 33
49 9E D9 C6 6A 83 9A C5 CD 2A C6 4D 47 57 B5 B4 8D 72 67 90 27 E6 6B EC 9D 26 C1 74 CD 26 D6 C9 00
FD CC 4A 9C 7B 0A F9 DB E0 B7 87 9B 53 F1 57 DB 66 43 E4 DA 82 C4 1F 5E D5 F4 B7 7A D6 0A C8 F2 EA
BB C8 06 7B D2 D2 63 9C D2 D5 99 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51
40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05
14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51
40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 05
14 51 40 05 14 51 40 05 14 51 40 05 14 51 40 08 7A 75 C5 37 1C 71 4F A4 C0 CE 71 40 AC 46 F1 AC AA 55
D0 30 EF B8 03 5C BE B3 F0 EB C3 5A CA B3 4F A7 C5 1C 87 FE 5A 46 30 6B AD EF 9A 30 2A 5C 53 29 49
AD 8F 13 D6 3E 06 B7 2F A3 DF AE 31 C2 4F 5C 16 AB F0 FF 00 C4 BA 3E E6 9F 4E 91 A3 1D 24 8F 91 F8 57
D5 18 00 D0 C0 30 C1 00 FD 45 4B A7 73 68 E2 26 8F 8D 66 82 48 5C 09 95 90 FF 00 BA 41 FD 6A 20 08
18 53 91 EB 5F 5A EA 9E 15 D0 F5 80 7E DB A7 40 ED FD E2 BC D7 0B AC 7C 12 D2 2E C1 7D 36 EA 5B 59
3A 80 4E 53 F2 AC DD 26 75 C3 1A 9A B4 8F 03 60 42 8D A7 BD 26 E3 5E 83 AB 7C 20 F1 2E 9E CF F6 68
E3 BB 40 3E FA 1C 0C 7D 2B 89 BB D1 35 4D 39 D9 6E EC 67 46 1E A8 71 F9 D4 F2 58 D2 35 A2 FA 94 F0
4F 26 9B 9C D3 D0 2B 30 5D DB 18 F5 0D 43 C6 D1 92 4A E7 DC 74 A8 B1 AF 32 64 65 DC 1D AA 71 4A 24
97 6E 36 03 EF 48 C4 06 0D 8F D2 9C 72 0E 2A 91 16 22 6D CC 79 38 A4 0A 7D 69 ED 8C 53 48 F9 73 41
4A C4 72 0A 88 02 4D 4E 46 53 03 AE 6A 36 E3 00 70 7B 9A 69 8D A2 33 4E 55 DE C1 31 92 7A 7D 68 65

03 83 92 7A 8C 77 AE EB E1 6F 84 25 F1 47 88 12 69 62 3F 60 B6 21 A4 62 38 63 E9 F5 AD 22 8E 4A B3 49 33 D9 BE 13 78 71 B4 2F 09 C5 73 70 B8 BA BB F9 DC 7A 2F 6A F4 0A 8E 38 D6 30 B1 22 85 8D 14 05 C7 F2 A9 6B 65 A1 C1 7B BB 89 DE 96 8A 28 10 51 45 14 00 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 45 00 25 14 B4 50 02 51 4B 50 50 04 D4 54 35 2D 00 2D 14 94 50 02 D1 50 D1 40 13 51 49 45 00 2D 14 95 15 00 4D 47 34 94 50 00 14 E0 82 73 46 38 F4 A2 8E F4 00 63 1C D4 17 36 76 D7 88 63 B9 82 39 50 F6 65 CD 4F 51 1A 4D 03 39 3D 5B E1 87 85 F5 73 99 2C 04 2D FD E8 4E DA E2 F5 4F 81 28 CC CF A6 6A 86 30 3E EA 4A 32 7F 3A F6 0A 3D 6A 5C 51 71 A9 25 D4 F9 B7 51 F8 4D E2 9B 28 DC F9 11 5C 46 BF C4 1C 0E 2B 93 BC D1 2F 2C 5C AD C4 3E 59 1D 46 EC D7 D5 5A D7 FC 83 A5 FF 00 76 BE 7D F1 67 FC 7C CD 51 25 63 B6 8C DC 96 A7 0A C1 73 D6 9A D9 C7 1D 28 7F BC 68 3F 72 B2 67 4C 46 96 55 EF DA A3 03 CC 3B 40 CB 1E D5 1B F6 A2 DB FE 3E 57 EA 28 88 54 7A 1D CF 86 7E 17 EB BE 28 9A 16 58 D6 DA C0 11 BA 62 E0 90 3D 31 EB 5F 48 F8 73 C3 B6 3E 18 D1 E2 D3 74 E8 82 44 83 2C 7B BB 77 26 B9 5F 85 BF F2 2F D7 A0 8A E8 89 E5 54 93 BD 86 AA 90 01 C7 7C E2 9F 50 D1 56 66 4D 45 43 45 00 4D 45 43 45 00 7F FF D9

7.2.3 Command used for combining the images:



7.3 APPENDIX C – RECOVERED DOCUMENTS RELEVANT TO PCAP 3 INVESTIGATION.

7.3.1 Final packet sent in TCP stream 234.

HTTP/1.1 200 OK

Date: Sun, 22 Oct 2023 16:49:30 GMT

Server: Apache/2.4.53 (Debian)

Last-Modified: Sun, 22 Oct 2023 16:47:55 GMT

ETag: "541-60850ddff6608-gzip"

Accept-Ranges: bytes

Vary: Accept-Encoding

Content-Encoding: gzip

Content-Length: 475

Connection: close

Content-Type: text/html

<div class='msgln'><i>User El Chapo has joined the chat.</i>
</div><div class='msgln'><i>User Narco Polo has joined the chat.</i>
</div><div class='msgln'>(12:46 PM) El Chapo: Good evening,

Narco Polo.
</div><div class='msgln'>(12:46 PM) Narco Polo: Who's on the line?
</div><div class='msgln'>(12:46 PM) El Chapo: Phoenix.
</div><div class='msgln'>(12:46 PM) Narco Polo: Where are you?
</div><div class='msgln'>(12:46 PM) El Chapo: I can't disclose that information, even to you.
</div><div class='msgln'>(12:46 PM) Narco Polo: Are you aware of the current scrutiny on El Chapo?
</div><div class='msgln'>(12:47 PM) El Chapo: Yes, I'm fully aware, However, they will never know it is me behind the shipment.
</div><div class='msgln'>(12:47 PM) Narco Polo: Regardless, we must exercise the highest level of secrecy. Be vigilant. I'd like to meet in 2nd November at 10 PM to plan the secret delivery and avoid any complications.
</div><div class='msgln'>(12:47 PM) El Chapo: At our usual rendezvous point?
</div><div class='msgln'>(12:47 PM) Narco Polo: Yes
</div><div class='msgln'>(12:47 PM) El Chapo: What day?
</div><div class='msgln'>(12:47 PM) Narco Polo: I already mentioned, stay sharp.
</div>

7.3.2 New message sent including coordinates.

<div class='msgln'>(12:53 PM) Narco Polo: 36.62575185817829 -117.08896804489794
</div>